CS544

Clarence Audate

Final Project

In this project, I analyze the **Student Habits Performance** dataset (`student_habits_performance.csv`) containing 1,000 students and 16 variables related to study habits, lifestyle, and exam performance. The first step is to import the data into R and clean/prepare it for analysis. The R code below demonstrates the import and initial preprocessing steps, with comments explaining each action:

1. Data Prep

```
# Load necessary libraries for analysis and plotting
library(ggplot2)     # for data visualization
library(dplyr)       # for data manipulation (e.g., sampling)
library(corrplot)    # for correlation plot

# Read CSV, treating the first row as column names
data <- read.csv(
  "student_habits_performance.csv",
  header = TRUE,          # first row are headers
  stringsAsFactors = FALSE # keep strings as character, not factors
)
sum(is.na(data))  # If 0, there are no missing values in the dataset

# Examine the structure of the data
str(data)  # Check data types and sample of each column
```

What it looks like:

```
> str(data)
'data.frame':     1000 obs. of  16 variables:
 $ student_id                  : chr  "S1000" "S1001" "S1002" "S1003" ...
 $ age                         : int  23 20 21 23 19 24 21 21 23 18 ...
 $ gender                      : chr  "Female" "Female" "Male" "Female" ...
 $ study_hours_per_day         : num  0 6.9 1.4 1 5 7.2 5.6 4.3 4.4 4.8 ...
 $ social_media_hours          : num  1.2 2.8 3.1 3.9 4.4 1.3 1.5 1 2.2 3.1..
 $ netflix_hours               : num  1.1 2.3 1.3 1 0.5 0 1.4 2 1.7 1.3 ...
 $ part_time_job               : chr  "No" "No" "No" "No" ...
 $ attendance_percentage       : num  85 97.3 94.8 71 90.9 82.9 85.8 77.7 100
 $ sleep_hours                 : num  8 4.6 8 9.2 4.9 7.4 6.5 4.6 7.1 7.5 ...
 $ diet_quality                : chr  "Fair" "Good" "Poor" "Poor" ...
 $ exercise_frequency          : int  6 6 1 4 3 1 2 0 3 5 ...
 $ parental_education_level    : chr  "Master" "High School" "High School"...
```

```
 $ internet_quality             : chr  "Average" "Average" "Poor" "Good" ...
 $ mental_health_rating         : int  8 8 1 1 1 4 4 8 1 10 ...
 $ extracurricular_participation: chr  "Yes" "No" "No" "Yes" ...
 $ exam_score                   : num  56.2 100 34.3 26.8 66.4 100 89.8 72.6
```

After loading, It's confirmed that **no missing values** are present because the sum of `NA` values is 0. The dataset includes both numerical and categorical features. The Key variables from this dataset include:

1. **Exam Score** (numerical, range 18.4–100, indicating student performance, main study of dataset),
2. **Study Hours per Day** (numerical, hours spent studying daily),
3. **Social Media Hours** and **Netflix Hours** (numerical, hours spent on these activities),
4. **Attendance Percentage** (numerical, class attendance rate),
5. **Sleep Hours** (numerical, hours of sleep per night),
6. **Exercise Frequency** (numerical, days exercising per week),
7. **Mental Health Rating** (numerical, self-rated 1–10),
8. **Gender** (categorical: Female, Male, Other),
9. **Part-Time Job** (categorical: Yes/No),
10. **Diet Quality** (categorical (ordinal) : Poor, Fair, Good),
11. **Parental Education Level** (categorical (ordinal) : None, High School, Bachelor, Master),
12. **Internet Quality** (categorical (ordinal): Poor, Average, Good), and
13. **Extracurricular Participation** (categorical: Yes/No).

```
 # Convert character columns into factors for proper categorical handling
> data$gender <- factor(data$gender)
> data$part_time_job <- factor(data$part_time_job)
> data$diet_quality <- factor(
+   data$diet_quality,
+   levels = c("Poor","Fair","Good"),  # define logical order
+   ordered = TRUE                      # make it an ordered factor
```

```
+ )
> data$parental_education_level <- factor(
+  data$parental_education_level,
+  levels = c("None","High School","Bachelor","Master"),
+  ordered = TRUE
+ )
> data$internet_quality <- factor(
+  data$internet_quality,
+  levels = c("Poor","Average","Good"),
+  ordered = TRUE
+ )
> data$extracurricular_participation <-
factor(data$extracurricular_participation)

> # Remove the student_id column—doesn't inform analysis
> data$student_id <- NULL
```

```
> summary(data)
  student_id             age           gender          study_hours_per_day social_media_hours
 Length:1000        Min.   :17.00   Length:1000        Min.   :0.00        Min.   :0.000
 Class :character   1st Qu.:18.75   Class :character   1st Qu.:2.60        1st Qu.:1.700
 Mode  :character   Median :20.00   Mode  :character   Median :3.50        Median :2.500
                    Mean   :20.50                      Mean   :3.55        Mean   :2.506
                    3rd Qu.:23.00                      3rd Qu.:4.50        3rd Qu.:3.300
                    Max.   :24.00                      Max.   :8.30        Max.   :7.200
 netflix_hours    part_time_job      attendance_percentage  sleep_hours    diet_quality
 Min.   :0.000   Length:1000        Min.   : 56.00        Min.   : 3.20   Length:1000
 1st Qu.:1.000   Class :character   1st Qu.: 78.00        1st Qu.: 5.60   Class :character
 Median :1.800   Mode  :character   Median : 84.40        Median : 6.50   Mode  :character
 Mean   :1.820                      Mean   : 84.13        Mean   : 6.47
 3rd Qu.:2.525                      3rd Qu.: 91.03        3rd Qu.: 7.30
 Max.   :5.400                      Max.   :100.00        Max.   :10.00
 exercise_frequency parental_education_level internet_quality   mental_health_rating
 Min.   :0.000      Length:1000              Length:1000        Min.   : 1.000
 1st Qu.:1.000      Class :character         Class :character   1st Qu.: 3.000
 Median :3.000      Mode  :character         Mode  :character   Median : 5.000
 Mean   :3.042                                                 Mean   : 5.438
 3rd Qu.:5.000                                                 3rd Qu.: 8.000
 Max.   :6.000                                                 Max.   :10.000
 extracurricular_participation   exam_score
 Length:1000                   Min.   : 18.40
 Class :character              1st Qu.: 58.48
 Mode  :character              Median : 70.50
                               Mean   : 69.60
                               3rd Qu.: 81.33
                               Max.   :100.00
```

We dropped `student_id` since it's just a unique identifier and doesn't contribute to analysis. The categorical variables were converted to factor type for proper handling in R. A quick `summary()` and `str()` check shows that the data types are correctly set (factors for categories, numeric for measurements) and that all values fall in reasonable ranges (e.g., no negative hours, exam scores between 0–100).

2. Data Analysis

To start, we examine the categorical variable and one numerical variable in isolation. We chose **Diet Quality** for the categorical analysis and **Exam Score** for the numerical analysis. These were selected because *diet quality* represents a key habit/lifestyle factor (with three levels allowing an interesting distribution view), and *exam score* is the primary performance outcome of interest (continuous metric of academic success). Examining these individually provides insight into the overall makeup of the dataset and the range of student performance.

Categorical Variable – Diet Quality: We first look at the distribution of diet quality among students. A frequency table (`table(data$diet_quality)`) shows how many students fall into each category (Poor, Fair, Good). We also compute proportions (`prop.table`) to see percentages.

```
> diet_counts <- table(data$diet_quality)  # raw counts by diet level
> print(diet_counts)

Poor Fair Good
 185  437  378

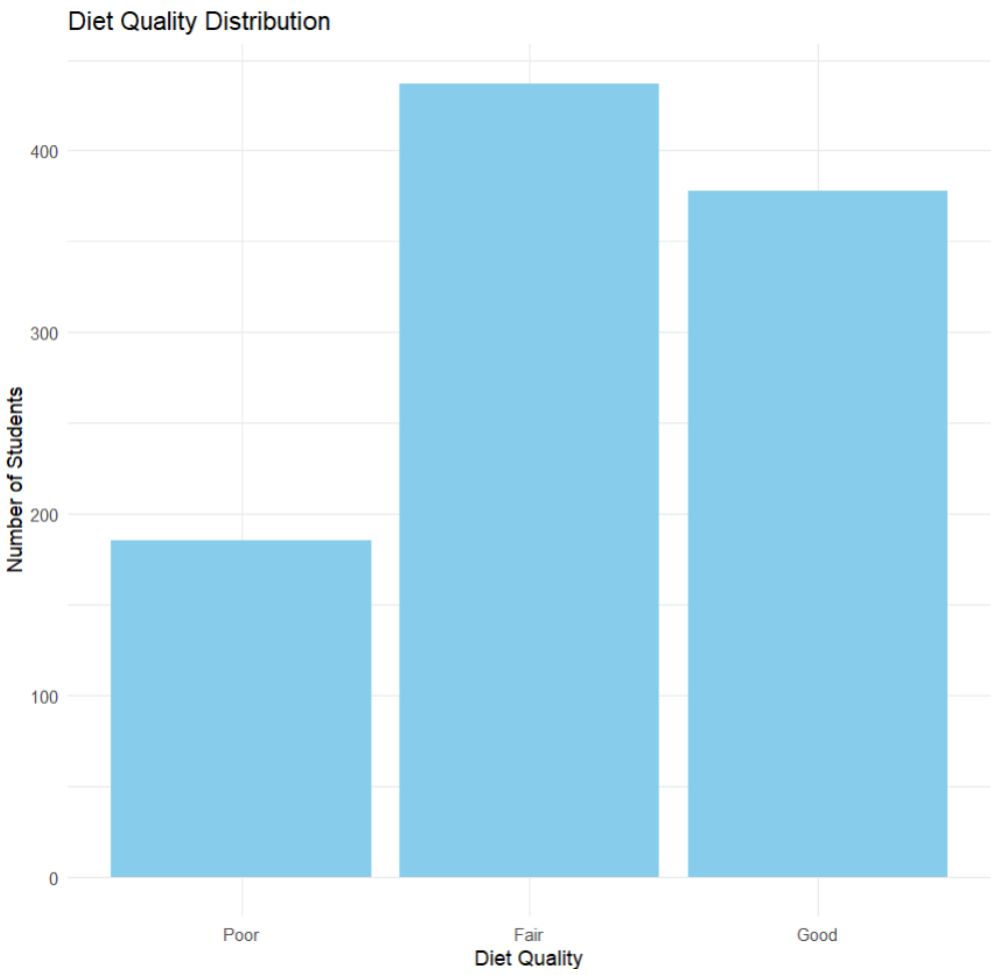
> df_diet <- as.data.frame(diet_counts)    # convert to 2-col data.frame
> colnames(df_diet) <- c("Diet", "Count")  # give clear column names
> print(df_diet)

  Diet Count
1 Poor   185
2 Fair   437
3 Good   378

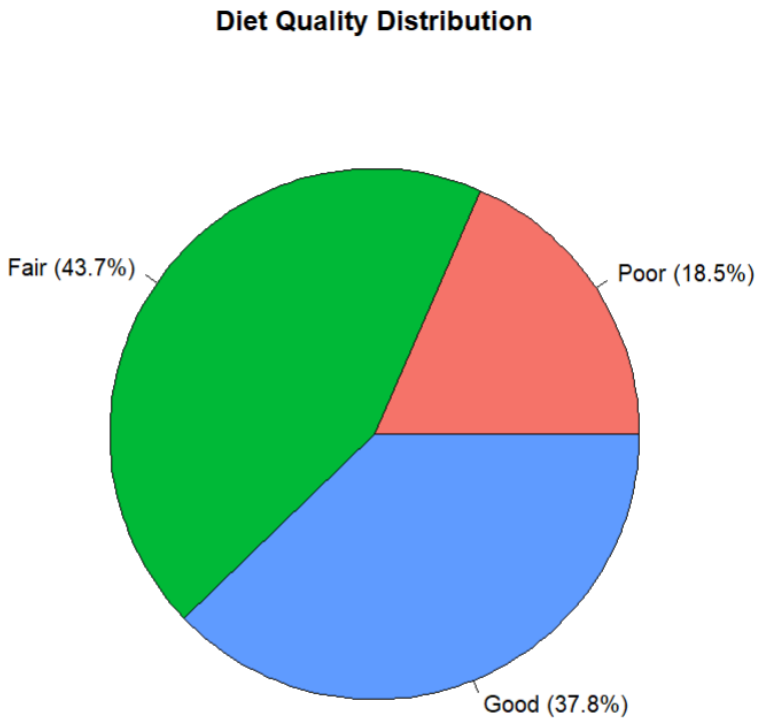
> prop.table(diet_counts) # show proportions

 Poor  Fair  Good
0.185 0.437 0.378

# Bar chart of diet distribution
> ggplot(data, aes(x = diet_quality)) +
+  geom_bar(fill = "skyblue") +           # bar height = count
+  labs(
+    title = "Diet Quality Distribution",
+    x     = "Diet Quality",
+    y     = "Number of Students"
+  ) +
+  theme_minimal()
```




```
#  Pie chart of diet distribution (base R)
> pie(
+  diet_counts,
+  labels = paste0(
+    names(diet_counts),
+    " (",
+    round(100 * diet_counts / sum(diet_counts), 1),
+    "%)"
+  ),
+  col = c("#F8766D", "#00BA38", "#619CFF"),
+  main = "Diet Quality Distribution"
+ )
```



Interpretation of diet quality:

Diet quality among students is mostly **Fair** or **Good**, with fewer students reporting a **Poor** diet. The bar chart shows that about 43.7% of students have a “Fair” diet and 37.8% have a “Good” diet, while roughly 18.5% report “Poor” diet quality. Therefore, the majority of students maintain at least a fair diet. This clear visualization of counts (shown on the y-axis) confirms that very few students fall into the lowest diet category.

Summary and SD of scores:

```
> summary(data$exam_score)  # min, 1st Q, median, mean, 3rd Q, max
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  18.40   58.48   70.50   69.60   81.33  100.00

> sd(data$exam_score)       # standard deviation
[1] 16.88856
```

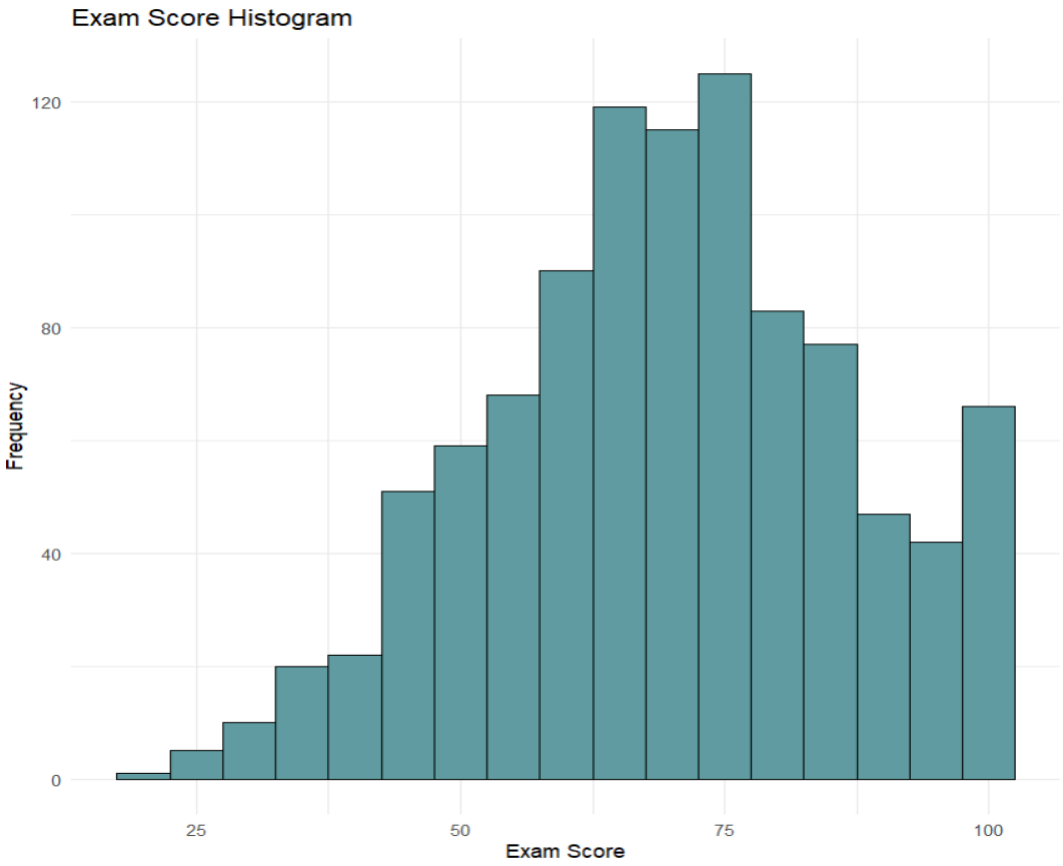
Interpretation of summary and sd of exam scores:

- **The mean (69.60)** is slightly below **the median (70.50)**, suggesting a slight left skew, but it's quite close — the distribution is roughly symmetrical.
- **The range** is wide: from **18.4 to 100**, indicating diverse performance levels.

- **The interquartile range (IQR)** is **81.33** - **58.48 = 22.85,** representing the **middle 50%** of scores.
- **A standard deviation** of **~16.89 s**hows that scores are fairly spread out around the mean.

Exam score visualizations:

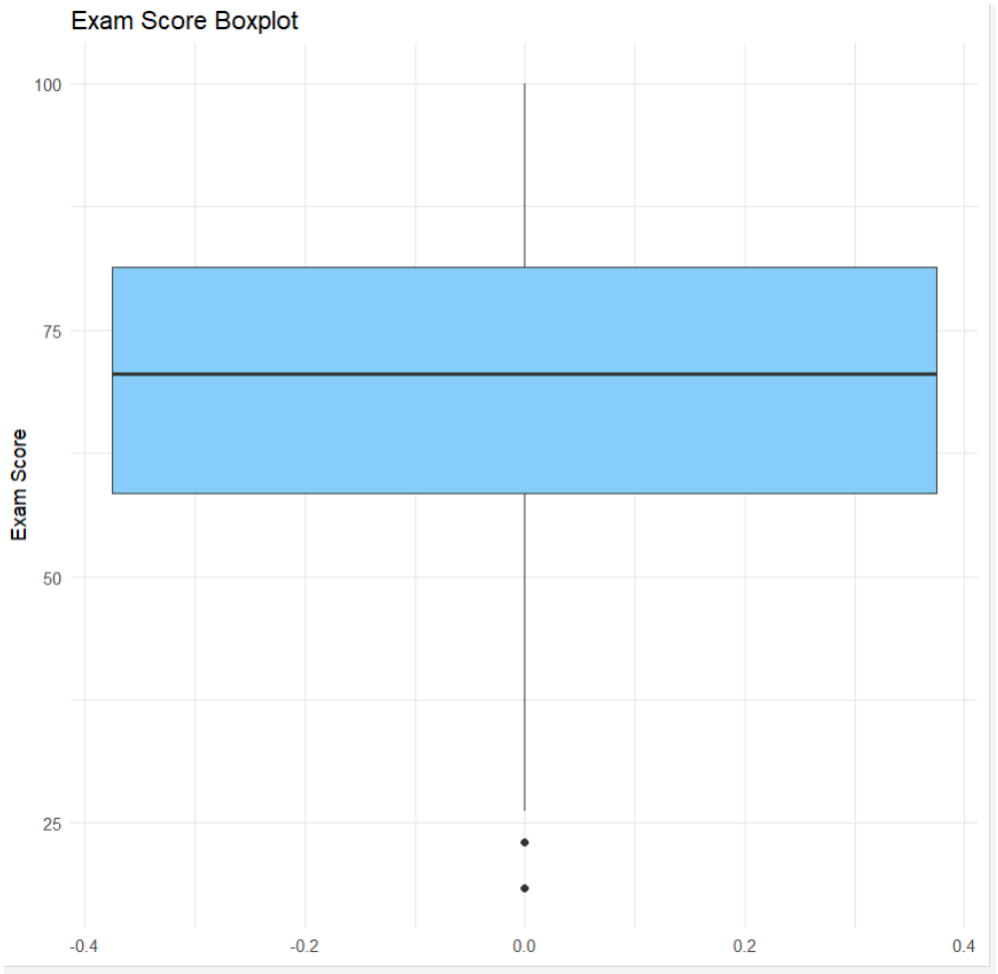
```
# Histogram of exam scores
ggplot(data, aes(x = exam_score)) +
  geom_histogram(
    binwidth = 5,           # 5-point bins
    fill     = "cadetblue",
    color    = "black"
  ) +
  labs(
    title = "Exam Score Histogram",
    x     = "Exam Score",
    y     = "Frequency"
  ) +
  theme_minimal()
```



Histogram Interpretation:

The histogram of Exam Score shows the overall distribution of student performance. The scores **appear skewed toward the higher** end since **many students scored in the 70–90 rang**e, with **a noticeable number achieving scores in the 90s up to 100**. There is **a tail toward the lower scores (20–50 range**), indicating a smaller group of students with poor exam performance. **The most frequent scores seem to cluster around the 60–80 range**, and there is a spike at 100. Overall, **the distribution is not perfectly symmetric**; it has a slight negative skew due to a few low-scoring outliers.

```
# Boxplot of exam scores
ggplot(data, aes(y = exam_score)) +
  geom_boxplot(fill = "lightskyblue") +
  labs(
    title = "Exam Score Boxplot",
    y     = "Exam Score"
  ) +
  theme_minimal()
```



Boxplot interpretation:

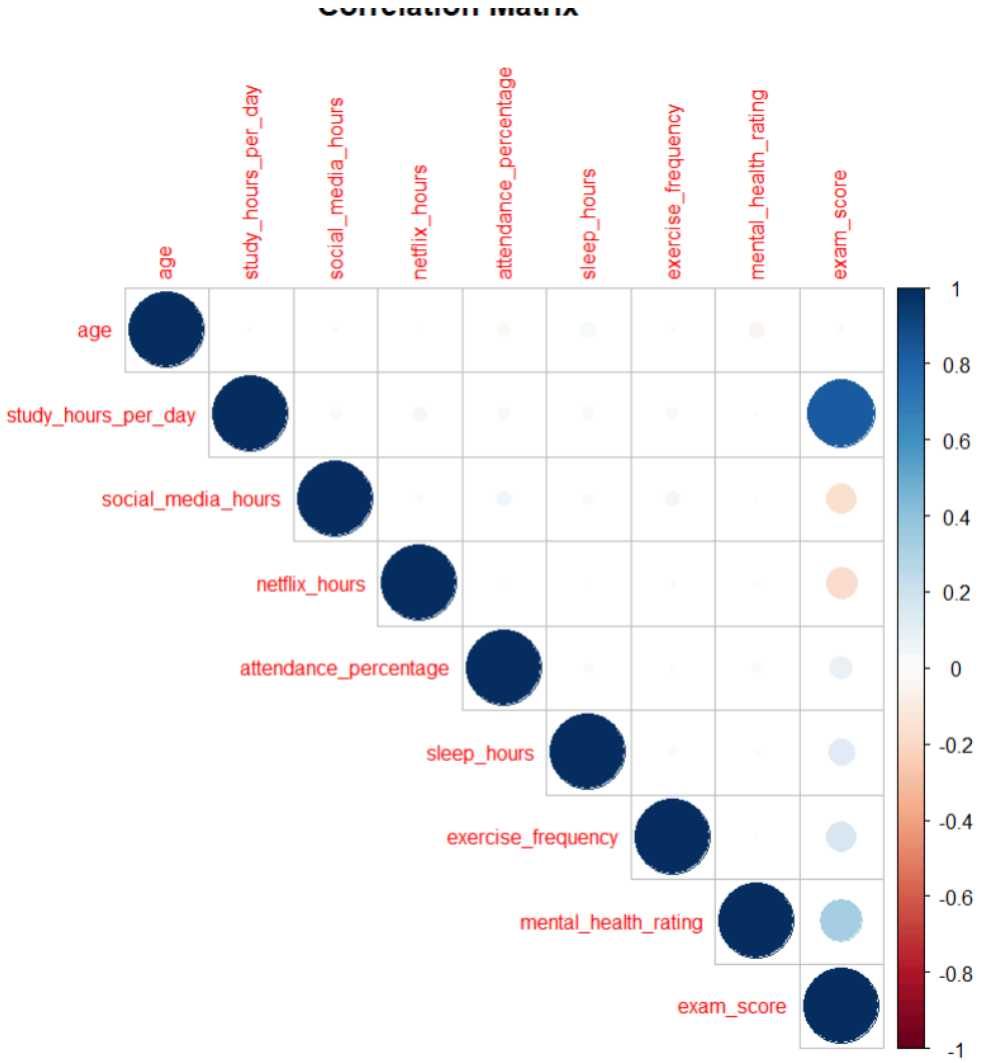
1. **Median (the thick line inside the box)**
    - That horizontal bar sits right around **70**, meaning half the students scored below ~70 and half scored above. **Typical performance** centers around 70.
2. **Interquartile Range (the box edges)**
    - The bottom of the box (Q1) is roughly **60**, and the top of the box (Q3) is roughly **82**.
    - So the middle 50% of the exam scores lie between about 60 and 82. **Half** of the students score between about **60 and 82.**
3. **Whiskers**
    - The lines extending up and down from the box reach to the most extreme data points **within 1.5 × IQR** of the quartiles.
    - Here, the upper whisker goes up to the maximum non-outlier score (100), and the lower whisker drops down to around **35**. **Most** scores drop between roughly **35 and 100.**
4. **Outliers (individual points below the lower whisker)**
    - A couple of isolated points down near **20–25**. Those are scores more than 1.5 × IQR below Q1—students who did unusually poorly compared to the rest. A **few outliers** scored very low.

Overall, the exam scores are reasonably spread with a few low-score outliers, but the **majority** of students **performed** in the **60–82** range**.**

Correlation between multiple variables:

```
# Identify numeric columns and compute correlation matrix
numeric_cols <- sapply(data, is.numeric)
cor_mat <- cor(data[, numeric_cols])

# Visualize correlation matrix
corrplot(
  cor_mat,
  type   = "upper",  # show only upper triangle
  tl.cex = 0.8,      # label size
  title  = "Correlation Matrix"
)
```



Interpretation of correlation matrix:

1. Strong positive relationships (dark blue, near 1.0)

- **Study Hours per Day** ↔ **Exam Score**: Very dark blue in the bottom-left corner—study time is the single strongest predictor of performance (r ≈ +0.83).
- **Mental Health Rating** ↔ **Exam Score**: Also a deep blue, though slightly lighter (r ≈ +0.32), indicating students who report better mental health tend to score higher.
- **Age** ↔ **Social Media Hours** and **Social Media** ↔ **Netflix Hours**: Dark blue between "age" & "social_media_hours" (older students use social media more?) and between "social_media_hours" & "netflix_hours" (those who use one tend to use the other).
  s

2. Moderate positives (medium blue)

- **Sleep Hours** ↔ **Mental Health**: Students who sleep more tend to rate their mental health higher.
- **Attendance Percentage** ↔ **Exam Score**: A very light blue suggesting attendance has a small positive correlation with exam outcomes.
- **Exercise Frequency** ↔ **Mental Health**: Light to medium blue , suggesting some link between more exercise and feeling mentally well.

3. Weak or negligible correlations (almost white)

- Many pairs (e.g., **Sleep** ↔ **Exam Score**, **Exercise** ↔ **Exam Score**, **Netflix** ↔ **Mental Health**) are nearly colorless, meaning very little linear relationship.

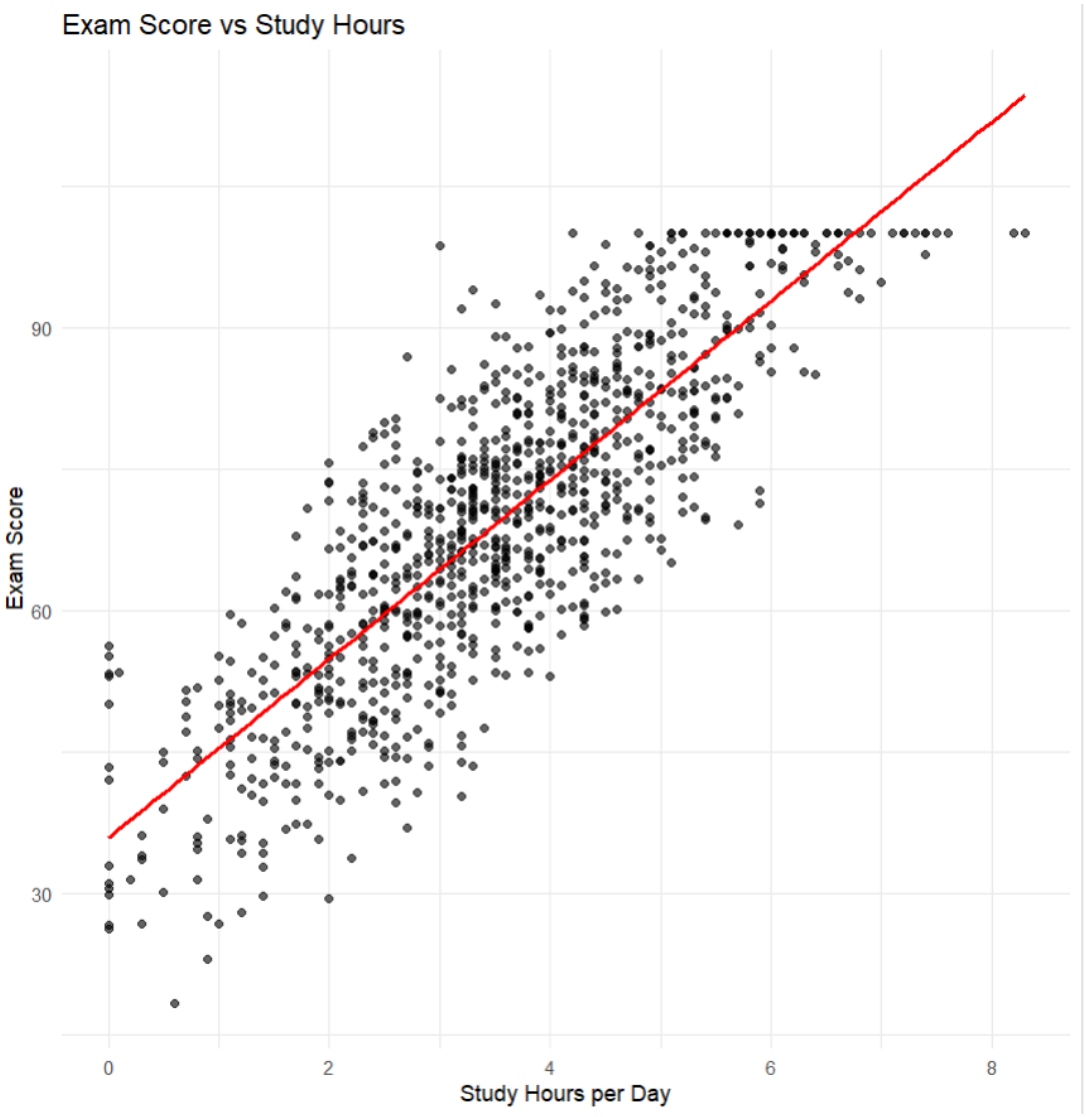
4. Weak negative relationships (pale red/orange)

- **Social Media Hours** ↔ **Exam Score** & **Netflix Hours** ↔ **Exam Score**: Slightly red, indicating that higher time on those leisure activities is marginally associated with lower scores.
- **Netflix Hours** ↔ **Sleep Hours**: A hint of red, suggests more streaming may slightly cut into sleep.

Overall, **Study habits** dominate: more study results in much higher scores. While **well-being factors** (mental health, sleep, exercise) have modest positive ties to performance. **Entertainment time** (social media, Netflix) correlates weakly negatively with scores. **Attendance** is only very lightly tied to exam outcomes in this sample. These results match the scatter-plot and regression findings with study time by far being the strongest driver of exam success, with other lifestyle factors playing smaller roles.

Exam score visualizations:

```
# Scatterplot with linear fit: study hours → exam score
ggplot(data, aes(x = study_hours_per_day, y = exam_score)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "red") + #add red regression line
  labs(
    title = "Exam Score vs Study Hours",
    x     = "Study Hours per Day",
    y     = "Exam Score"
  ) +
  theme_minimal()
```



Scatterplot with regression line interpretation of exam score and study hours:

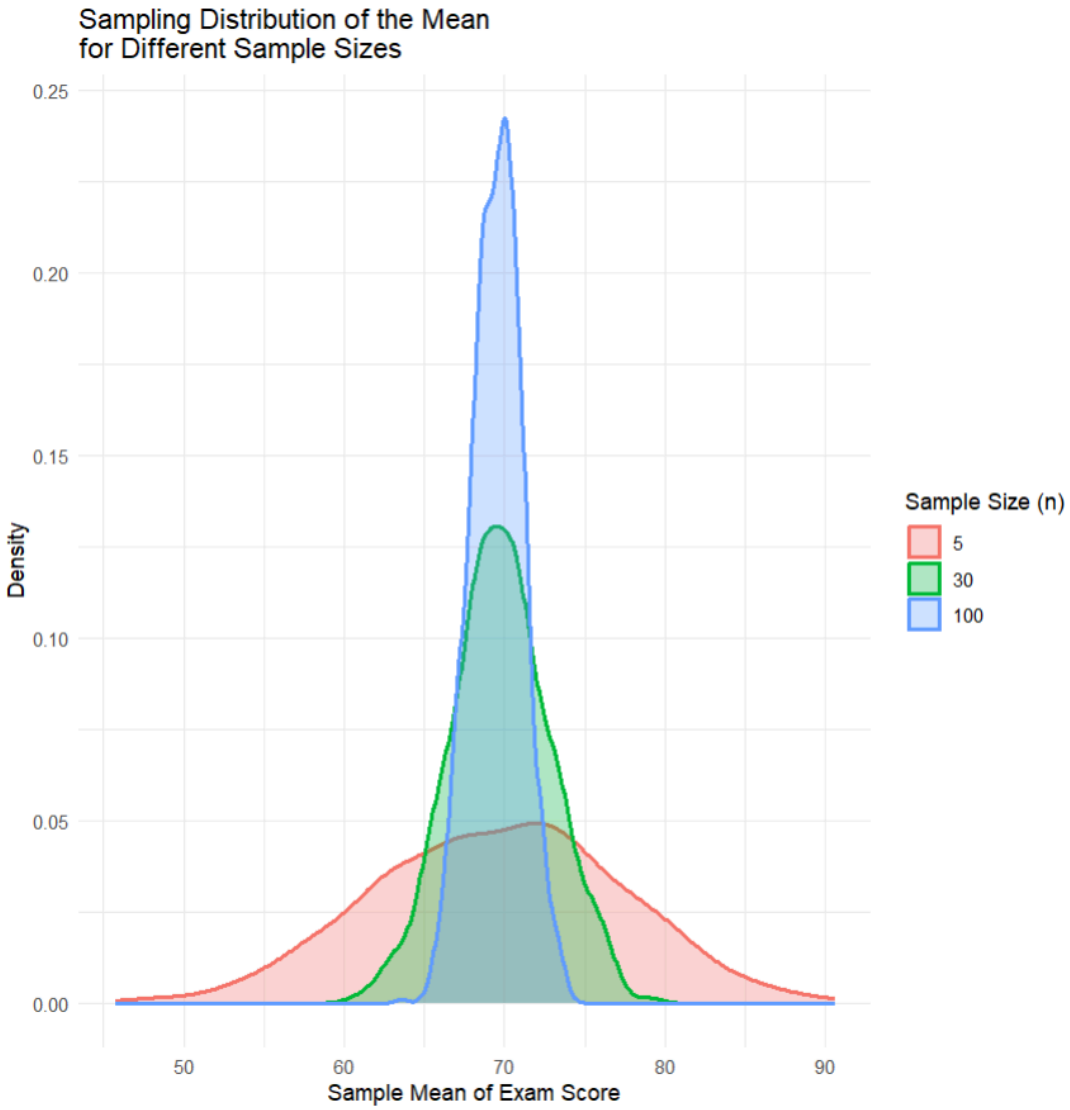
The scatter plot confirms **a clear positive linear relationship between study time and exam performance**. Each point represents a student. We see that **students who studied very little (near 0 hours/day) mostly scored quite low (many points in the 20–50 score range for 0–1 study hour)**. In contrast, students who dedicate more time to studying **(e.g., 5–8 hours per day) tend to cluster at higher exam scores (80–100 range)**. **The red regression line emphasizes this trend**, ascending steeply – for each additional hour studied per day, exam scores tend to increase significantly. The points are reasonably tight around the line, indicating consistent improvement with more study. There is some variability (not every student with low study hours fails, nor does every high-study student score perfectly), but the upward trend is clear.

The **Central Limit Theorem (CLT)** is one of the cornerstones of statistics. In plain English, it says:

> **"If you take sufficiently large random samples from any population (regardless of its original distribution), the distribution of the sample means will tend to be *approximately Normal (bell-shaped)*, with mean equal to the population mean and standard deviation equal to the population standard deviation divided by √n.**

Why it matters

It justifies using the **Normal (bell-curve) distribution**, specifically, the standard Normal, to build intervals around a sample mean or to test claims about a population mean, even when the original data aren't Normal, as long as your sample is large enough (commonly n ≥ 30).



CLT Interpretation:

- **Larger samples result in more precise estimates:** As n increases, the sampling distribution of x becomes both *more normal in shape* and *less spread out* (variance decreases by a factor of 1/n).

- With n=100, the sample mean from just 100 students is very likely to lie within a couple of points of the true mean. With n=5, you'd get wildly different x each time you sample.

Sampling Methods:

```
> print(mean(srs$exam_score)) #simple random sampling

[1] 72.026

> print(mean(strat$exam_score)) #stratified sampling

[1] 69.19

> print(mean(sys$exam_score)) #systematic sampling

[1] 72.031

> print(mean(data$exam_score))  #population mean

[1] 69.601
```

Sampling methods interpretation:

- **Sampling variability**
  Each sample of 100 students gives a slightly different average just by chance. None of the sample means exactly equals the true population mean (69.60), because when you draw a subset, you'll inevitably over- or under-represent certain students.

- **Unbiasedness of the mean**
  Although any **one** sample might be above or below the true mean, over many repetitions the **average of those sample means** would be very close to 69.60. That's why we call the sample mean an **unbiased estimator** of the population mean.

- **Comparison of methods**

    - The **stratified sample** happened to produce a mean (69.19) very close to 69.60—reflecting its design to mirror the population's gender breakdown.

    - The **simple random** and **systematic** samples each came out at about 72.03, slightly higher than the true mean. That's just random luck in those draws—they happened to include more higher-scoring students.

These results illustrate **sampling error**: any single sample's mean can deviate from the true mean. The amount of deviation (here up to ~2.4 points) is typical for a sample of 100 from this population. If you repeated the sampling many times, the **distribution of those sample means** would center on 69.60 (the population mean) and have a spread exactly as the Central Limit Theorem predicts.

Jobs and extra curricular activities:

```
#create dataframe table

> prop_df <- as.data.frame.matrix(round(prop_job_extra * 100, 1))  # convert to
% and round

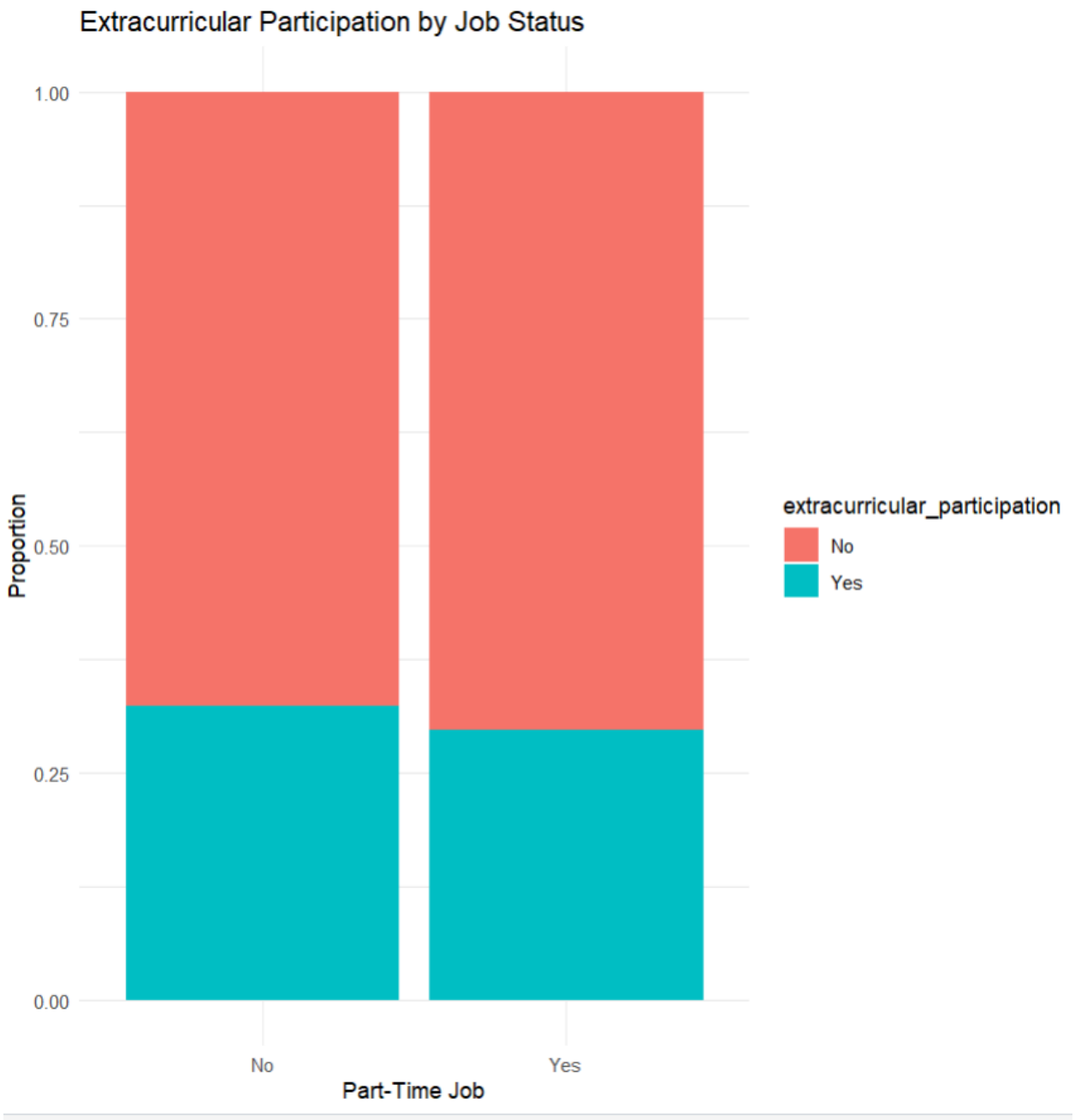
> colnames(prop_df) <- c("No Extracurricular (%)", "Yes Extracurricular (%)")

> rownames(prop_df) <- c("No Part-Time Job", "Has Part-Time Job")

> kable(prop_df, caption = "Proportion of Extracurricular Participation by
Part-Time Job Status")
```

Table: Proportion of Extracurricular Participation by Part-Time Job Status

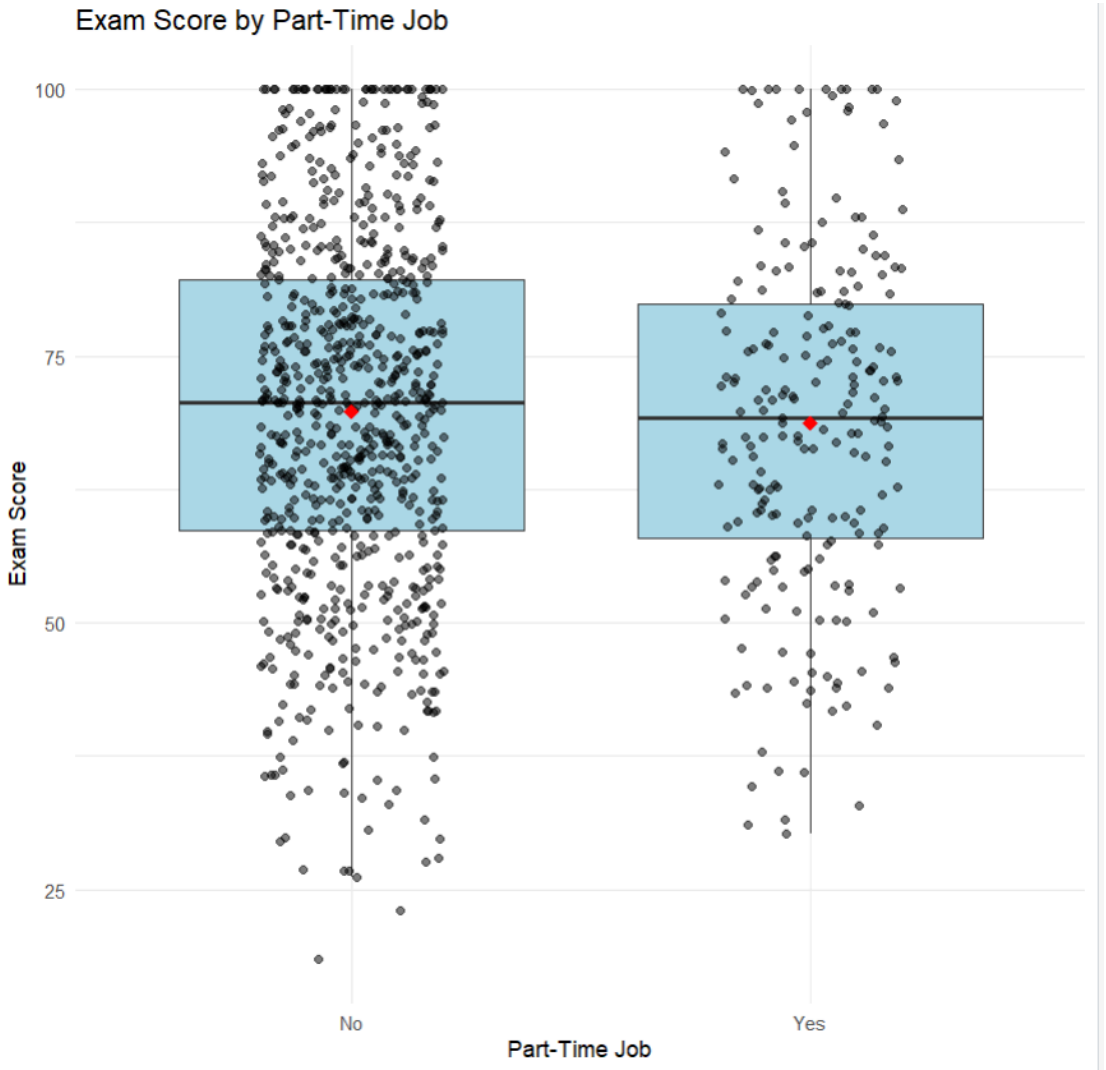
	No Extracurricular (%)	Yes Extracurricular (%)
No Part-Time Job	67.6	32.4
Has Part-Time Job	70.2	29.8



Relationship: Students **without** a part-time job are slightly **more likely** to do extracurricular activities (32.4% vs. 29.8%). Also, those **with** a job have a marginally lower participation rate (70.2% do **not** participate). These numbers have very little impact on the exam score overall.

More Visualizations between variables:

Exam score by part time job box plot:

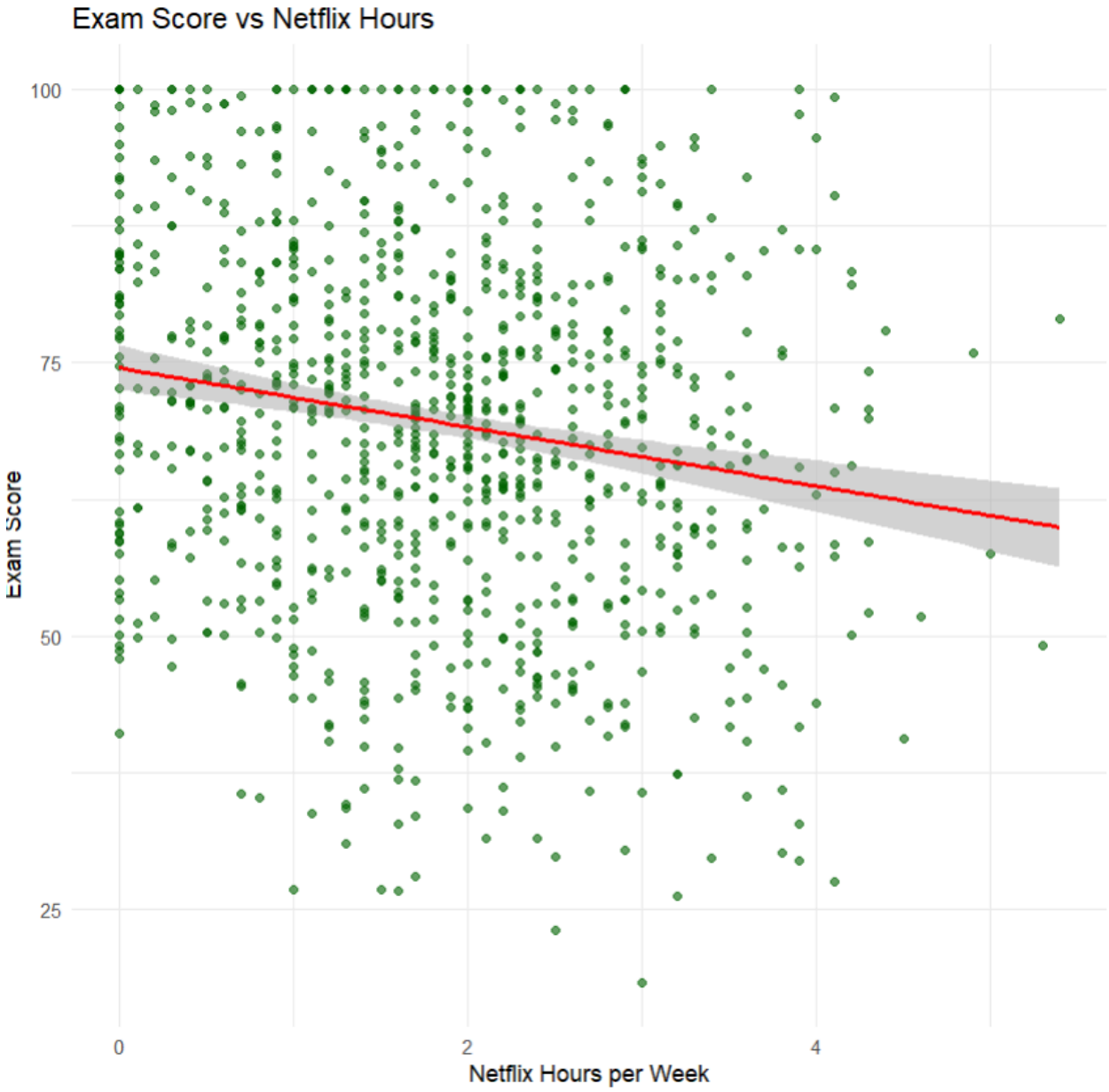


Interpretation:

Students **without** a part-time job score **very slightly higher** on average and have a marginally higher median than those **with** a job.

However, the distributions **overlap heavily**, indicating that having a part-time job is **not a strong determinant** of exam performance in this sample; other factors such as study hours play a much larger role.

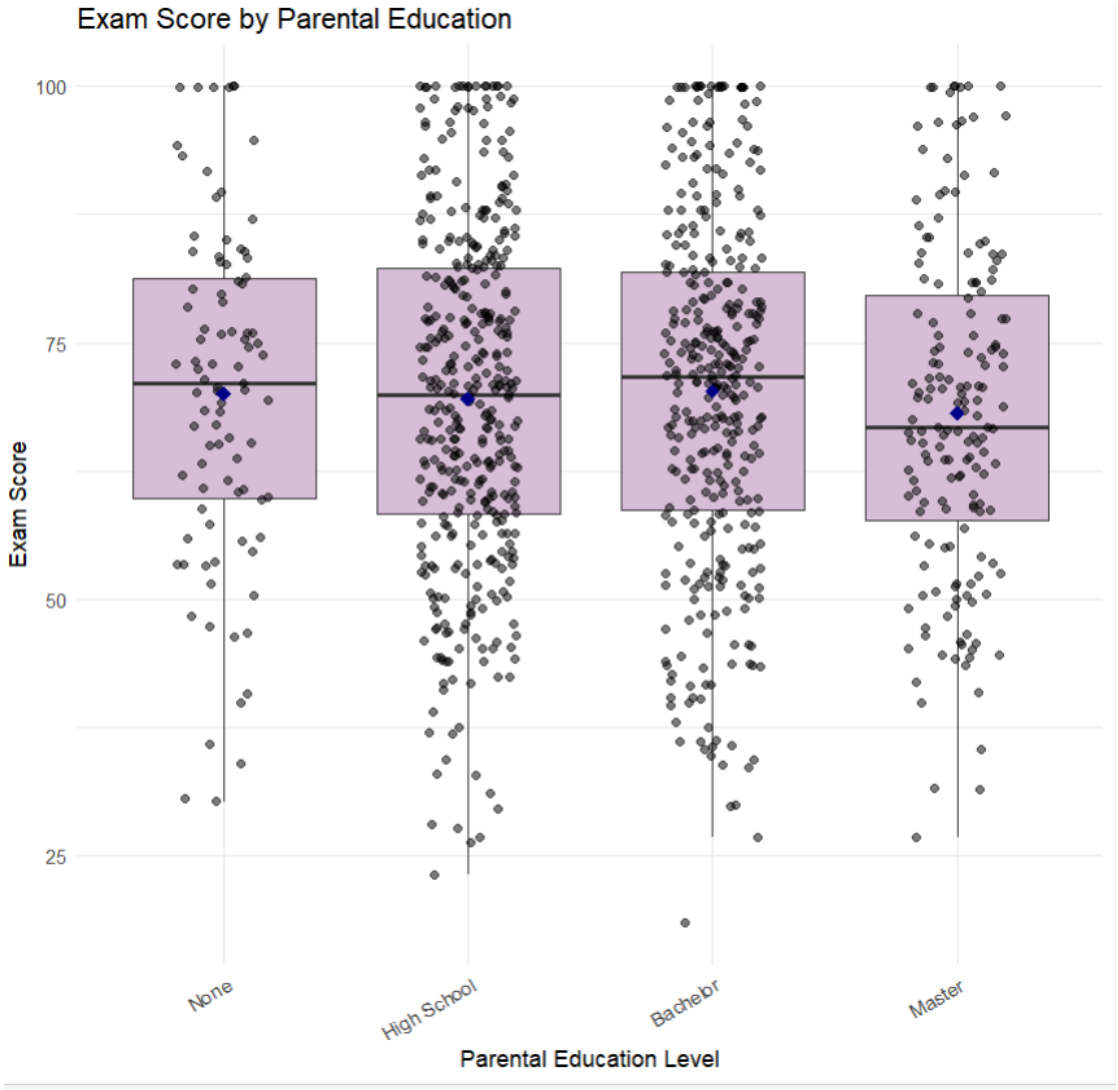
Exam score vs netflix hours:



Interpretation:

The red regression line slopes gently downward from left to right, indicating a small negative relationship: on average, students who spend more time on Netflix tend to score slightly lower on exams. However, the cloud of points shows that many high scoring students do watch Netflix, and many low scoring students watch little or none, so it's far from a definitive driver of performance

Parent's education level vs exam score:



Interpretation:

Slight advantage for "Bachelor" parents: Students whose parents hold a Bachelor's degree have the highest median and mean exam scores, suggesting a small positive association.

Surprising dip for "Master" parents: The Master-educated group's median and mean are a little bit lower this may reflect a smaller sample size or other confounding factors (e.g. socio-economic differences not captured here).

Overall weak effect: The boxes overlap heavily, and the ranges are almost the same. That tells us **parental education level is at best a modest predictor** of exam performance in this dataset, much weaker than the effect of study hours.

3. Summary

Variable Selection: We chose **Exam Score** as the primary numerical variable because it directly measures academic performance, the outcome of greatest interest. Analyzing exam scores allowed us to understand the overall performance distribution of students and is crucial for relating performance to various habits. For the categorical analysis, we selected **Diet Quality** as it represents a key aspect of student lifestyle (health/nutrition) and has three meaningful levels (Poor/Fair/Good). This choice provided insight into the general well-being of the student population and could potentially relate to performance or other habits. The diet quality variable was chosen for its clear categorical gradation and relevance to student health, with the understanding that its pros are an intuitive ordinal scale, while a con could be that it's self-reported and subjective

Univariate Analysis Findings: The examination of single variables revealed the makeup of our dataset. The **Diet Quality** distribution showed that most students report Fair or Good diets (~81% combined), with only ~19% having Poor diets. This suggests that a large majority have at least moderate diet habits, which is positive. On the performance side, the **Exam Score** distribution was centered around 70, with a broad range from 18 to 100. The histogram indicated a slight left tail (a few very low scorers) and a concentration of students in the middle-to-high performance range. The boxplot confirmed median performance around 70 and a few low outliers. Together, these indicate that while many students perform well, there is a minority struggling (scores <40), and overall performance is fairly spread out (standard deviation ~16.9 points).

Multivariate Analysis Findings: Exploring relationships between variables resulted in some strong and also some weak associations. The **most impactful finding** was the *strong positive correlation between* ***study habits and performance***: students who devoted more hours to study per day tended to score much higher on exams. This was evidenced by a correlation between study hours and exam scores and a clear upward trend in the scatterplot.

We also found that **mental health rating correlated moderately with exam scores** , suggesting a link between well-being and performance because students feeling mentally healthier may concentrate better or manage stress effectively, reflecting in their grades. However, relationships between performance and leisure activities were relatively weak, for example, **social media and Netflix time had small negative correlations with exam score**, implying that high usage of these might mildly detract from study time or focus. Additionally, I checked numeric-categorical relationships: for example, I saw **students with Poor diet had slightly lower exam scores on average** than those with Fair/Good diets (median ~67 vs ~70), hinting that diet could have a minor impact on academic energy or focus. However, these differences were not as pronounced as the study hour effect. In summary, the multivariate analysis pinpointed study hours as a crucial factor, with mental health as a noteworthy

secondary factor, whereas variables like sleep, exercise, and entertainment habits did not show substantial impacts on exam outcomes.

Overall Key Insights: Combining all parts of the analysis, we can conclude a few key points. **Study habits are crucial** – encouraging students to spend more focused time studying is likely to result in better exam results, as evidenced by the strong correlation observed. **Balanced life factors like mental health and perhaps diet have noticeable but smaller influences** on performance; supporting student wellness could have academic benefits, although it's not as immediately quantifiable as study time. **Leisure activities (social media/Netflix) did not show a dramatic impact** on scores in this data, which might imply that moderate use of such media is not catastrophic to performance, though excessive use should be monitored as it did correlate negatively to a slight extent. **Most students manage a fair diet and a significant number participate in part-time work or extracurricular activities**, indicating a diverse range of commitments and availability to study more hours. We saw that working a job might slightly reduce extracurricular involvement, hinting at time trade-offs, but many students balance both.

These insights could be valuable for educators and students alike: focusing on effective study practices and maintaining mental health can markedly improve academic performance, while moderate extracurricular engagement and even part-time work need not be detrimental if managed well.